Cloud Computing Cybersecurity Issues: Threats, Vulnerabilities, Attacks and Protection Mechanisms

Cloud computing has transformed how governments, universities, financial institutions, businesses, and research organizations store, process, and share information. However, the migration of IT infrastructure, applications, databases, and services to cloud environments introduces significant cybersecurity challenges.

Cloud cybersecurity is particularly important because organizations no longer maintain complete physical control over their computing infrastructure. Data, applications, virtual machines, networks, storage, and security controls may be distributed across multiple data centres, cloud providers, and geographical locations.

The major concern is that a security failure in one component of a cloud environment may compromise the confidentiality, integrity, and availability of the entire cloud-based service.

This discussion covers the major cloud computing cybersecurity issues, attack mechanisms, real-world incidents, AI/ML-based security mechanisms, and a proposed cloud security architecture.

2. Major cybersecurity issues in cloud computing

The following are some of the most significant cloud cybersecurity threats.

The Cloud Security Alliance's 2026 report identifies 11 leading cloud security threats based on industry surveys. Inadequate Identity and Access Management ranks first, while AI-enhanced attacks and AI system compromise have emerged as new major concerns.

2.1 Cloud misconfiguration

High risk

Misconfiguration is one of the most common causes of cloud security incidents.

Examples:

- Publicly accessible cloud storage buckets.
- Databases exposed to the internet.
- Incorrect firewall and security group rules.
- Unprotected virtual machines.
- Incorrect access permissions.
- Exposed administrative interfaces.
- Insecure default configurations.

Example attack scenario:

1. An organization stores confidential research data in cloud storage.
2. The storage bucket is accidentally configured for public access.
3. An attacker discovers the exposed storage location.
4. The attacker downloads confidential files.
5. Sensitive information is disclosed without compromising the cloud provider itself.

Security mechanisms:

- Cloud Security Posture Management (CSPM).
- Automated configuration auditing.
- Infrastructure as Code (IaC) security scanning.
- Continuous compliance monitoring.
- Automated remediation of dangerous configurations.

2.2 Identity and Access Management (IAM) attacks

Critical risk

Identity and Access Management controls who can access cloud resources and what operations they can perform.

Attackers frequently target user credentials, administrator accounts, API keys, access tokens, and service accounts.

Major vulnerabilities:

- Weak passwords.
- Absence of Multi-Factor Authentication (MFA).
- Stolen credentials.
- Excessive user privileges.
- Poorly managed service accounts.
- Compromised API keys.
- Privilege escalation.
- Session hijacking and token theft.

Example:

An attacker obtains the credentials of a cloud administrator through phishing. The attacker then logs into the cloud console and accesses virtual machines, databases, and storage systems.

Protection mechanisms:

1. Multi-Factor Authentication.
2. Role-Based Access Control (RBAC).
3. Attribute-Based Access Control (ABAC).
4. Privileged Access Management (PAM).
5. Identity federation.
6. Short-lived access tokens.
7. Continuous identity monitoring.

8. Zero Trust Architecture (ZTA).

A particularly important principle is Least Privilege Access: users should receive only the permissions necessary to perform their assigned responsibilities.

2.3 Insecure APIs and interfaces

Cloud environments depend heavily on APIs for communication between applications, databases, virtual machines, storage, and cloud management services.

An insecure API can provide attackers with an entry point into the cloud infrastructure.

Common API vulnerabilities:

- Broken authentication.
- Broken authorization.
- API key exposure.
- Excessive data exposure.
- Injection attacks.
- Inadequate input validation.
- Unrestricted API access.
- Rate-limit bypass.

Example attack:

An attacker discovers an API endpoint that retrieves student information from a university cloud database. Due to inadequate authorization checks, the attacker changes a student ID in the API request and accesses another student's personal information.

Protection mechanisms:

- API Gateway.
- Web Application Firewall (WAF).
- OAuth 2.0 and OpenID Connect.
- Mutual TLS (mTLS).
- API authentication and authorization.
- Rate limiting.
- Input validation.
- API security testing.

2.4 Data breaches and unauthorized data access

Critical risk

A cloud data breach occurs when an unauthorized individual gains access to confidential or sensitive information.

Cloud databases may contain:

- Personal identification information.

- Financial records.
- Medical information.
- Government records.
- University research data.
- Intellectual property.
- Authentication credentials.

Attack mechanisms:

- Stolen credentials.
- SQL injection.
- Misconfigured cloud storage.
- Insider threats.
- Exploitation of vulnerable applications.
- Compromised third-party services.

Protection mechanisms:

Security mechanism	Purpose
AES-256 encryption	Protect data at rest
TLS 1.3	Protect data in transit
Database encryption	Protect database contents
Tokenization	Replace sensitive values with tokens
Data Loss Prevention (DLP)	Detect and prevent unauthorized data transfers
Data Access Governance	Control access to sensitive data
Data classification	Identify different levels of sensitivity

2.5 Account hijacking and credential theft

Account hijacking occurs when attackers obtain control over legitimate cloud accounts.

Common techniques:

- Phishing.
- Credential stuffing.
- Brute-force attacks.
- Password spraying.
- Session cookie theft.
- OAuth token theft.
- Malware-based credential harvesting.

Consequences:

- Unauthorized access to cloud applications.
- Theft of confidential data.
- Creation of unauthorized virtual machines.
- Modification of security configurations.

- Installation of ransomware.
- Financial losses through unauthorized resource consumption.

Protection:

- Phishing-resistant MFA.
- Passwordless authentication.
- Conditional access policies.
- Continuous session validation.
- Detection of unusual login locations and devices.
- Automated account suspension following suspicious activity.

2.6 Cloud ransomware attacks

Ransomware is malicious software that encrypts files or otherwise makes organizational data unavailable and demands payment for recovery.

Cloud environments can be affected through:

- Compromised cloud administrator accounts.
- Infected virtual machines.
- Compromised endpoints synchronized with cloud storage.
- Exploited application vulnerabilities.
- Compromised backup accounts.
- Abuse of cloud storage APIs.

Typical attack lifecycle:

Initial compromise

Phishing / stolen credentials / vulnerable application

Privilege escalation

Obtaining higher cloud permissions

Lateral movement

Accessing additional cloud resources

Data exfiltration and encryption

Stealing or encrypting organizational data

Extortion and service disruption

Ransom demand, operational disruption and possible data disclosure

Protection mechanisms:

- Immutable backups.
- Offline backups.
- Endpoint Detection and Response (EDR).
- Network segmentation.
- Privileged account protection.
- Ransomware behaviour detection.
- Recovery testing.
- Separate backup administration accounts.

2.7 Distributed Denial of Service (DDoS)

DDoS attacks attempt to make cloud-hosted services unavailable by overwhelming them with malicious traffic or requests.

Types of DDoS attacks:

Attack type	Mechanism
Volumetric	Flooding networks with massive traffic
Protocol-based	Exploiting network protocol weaknesses
Application-layer	Overloading HTTP, DNS or application services
DNS amplification	Using DNS infrastructure to amplify traffic
HTTP flood	Sending excessive HTTP requests

Consequences:

- Website unavailability.
- Disruption of online education.
- Financial transaction failures.
- Service degradation.
- Increased cloud infrastructure costs.

Protection:

- Cloud DDoS protection services.
- Web Application Firewall.
- Rate limiting.
- Traffic filtering.
- Content Delivery Network (CDN).
- Auto-scaling with spending controls.
- Network anomaly detection.

2.8 Insider threats

Insider threats originate from individuals who already have legitimate access to cloud systems.

These may include:

- Employees.
- System administrators.
- Database administrators.
- Developers.
- Contractors.
- Third-party service personnel.

Types:

1. Malicious insider: deliberately steals or destroys information.
2. Negligent insider: accidentally exposes confidential data.
3. Compromised insider: an employee's account is taken over by an attacker.

Example: A cloud administrator downloads confidential research information and transfers it to an external storage account.

Protection mechanisms:

- User Behaviour Analytics (UBA).
- Privileged Access Management.
- Data Loss Prevention.
- Separation of duties.
- Audit trails.
- Continuous access monitoring.
- Regular access reviews.

2.9 Insecure virtualization and hypervisor vulnerabilities

Virtualization is fundamental to cloud computing. Multiple virtual machines may operate on the same physical server.

The hypervisor manages the underlying physical resources and isolates virtual machines.

A vulnerability in the hypervisor may threaten the isolation between virtual machines.

Major threats:

- Hypervisor exploitation.
- VM escape.
- VM-to-VM attacks.
- Insecure VM images.
- Virtual machine sprawl.
- Snapshot theft.
- Unauthorized VM cloning.
- Side-channel attacks.

VM escape: An attacker compromises a virtual machine and attempts to cross the isolation boundary to interact with the hypervisor or host system.

Protection:

- Hypervisor security patching.
- Strong VM isolation.
- Secure VM images.
- Hardware-assisted virtualization security.
- Virtual machine integrity monitoring.
- Network micro-segmentation.

2.10 Supply chain attacks

Cloud services depend on many third-party components, including software libraries, APIs, plugins, development tools, managed services, and external vendors.

An attacker may compromise a third-party component and use it as an entry point into an organization's cloud environment.

Examples:

- Malicious software packages.
- Compromised CI/CD pipelines.
- Vulnerable third-party libraries.
- Compromised software updates.
- Exposed vendor credentials.
- Insecure managed service integrations.

Protection mechanisms:

- Software Bill of Materials (SBOM).
- Software Composition Analysis (SCA).
- Dependency vulnerability scanning.
- Secure CI/CD pipelines.
- Vendor security assessments.
- Code signing.
- Third-party access restrictions.

2.11 Advanced Persistent Threats (APTs)

APTs are sophisticated attacks in which threat actors maintain unauthorized access to an organization's systems for an extended period.

Cloud infrastructure is an attractive target for espionage, intellectual property theft, and strategic information collection.

Typical attack activities:

- Initial compromise through phishing.
- Credential theft.
- Privilege escalation.
- Lateral movement.
- Persistence through cloud identities.
- Data exfiltration.

- Concealment of malicious activities.

Protection mechanisms:

- Threat intelligence.
- Security Information and Event Management (SIEM).
- Extended Detection and Response (XDR).
- Endpoint Detection and Response (EDR).
- Cloud Detection and Response (CDR).
- Behavioural analytics.
- Threat hunting.

2.12 Cloud visibility and logging deficiencies

Organizations may not have adequate visibility into their cloud environments, particularly when resources are distributed across multiple regions and providers.

Problems include:

- Incomplete audit logs.
- Unmonitored cloud resources.
- Lack of centralized security monitoring.
- Inconsistent log retention.
- Insufficient network visibility.
- Delayed detection of suspicious activity.

Protection:

- Centralized logging.
- SIEM integration.
- Cloud-native audit logging.
- Continuous monitoring.
- Security dashboards.
- Automated incident alerting.

2.13 Cryptojacking and unauthorized resource consumption

Cryptojacking occurs when attackers secretly use cloud computing resources to mine cryptocurrency.

Attackers may compromise cloud accounts, virtual machines, containers, or exposed computing services.

Consequences:

- Increased cloud bills.
- Reduced processing capacity.
- Performance degradation.
- Unauthorized resource consumption.
- Potential compromise of other cloud services.

Protection:

- CPU/GPU utilization monitoring.
- Unusual resource consumption detection.
- Cloud cost anomaly detection.
- Runtime malware detection.
- Resource quotas.
- Automated shutdown of suspicious instances.

2.14 Cloud-specific AI security threats

AI introduces two dimensions of cloud security risks: attackers can use AI to improve their attacks, and attackers can directly compromise AI systems hosted in the cloud.

Threat	Description
AI-enhanced phishing	Highly convincing targeted phishing
Automated vulnerability discovery	AI-assisted identification of weaknesses
Prompt injection	Manipulation of cloud-hosted LLM applications
Data poisoning	Manipulation of training data
Model theft	Unauthorized copying of proprietary models
Model extraction	Reconstructing model behaviour through queries
Model supply chain attacks	Compromising model dependencies or artifacts
AI agent compromise	Abusing agent permissions, tools and credentials
Training data leakage	Exposure of confidential training information

Protection mechanisms:

- AI model access control.
- Secure model registries.
- Training data validation.
- Prompt injection protection.
- Model integrity verification.
- AI activity monitoring.
- Runtime guardrails.
- Human approval for sensitive agent actions.

The 2026 CSA report specifically highlights AI-enhanced attacks and AI system compromise as two newly prominent cloud security threats.

2.15 **Summary of cloud cybersecurity threats**

The following table summarizes the major issues and corresponding protection technologies.

No.	Cybersecurity issue	Major security solution
1	Cloud misconfiguration	CSPM

2	IAM attacks	MFA, RBAC, ZTA
3	Insecure APIs	API Gateway, WAF
4	Data breaches	Encryption, DLP
5	Account hijacking	MFA, UEBA
6	Ransomware	EDR, immutable backup
7	DDoS	CDN, DDoS protection
8	Insider threats	PAM, UEBA
9	Hypervisor attacks	Hypervisor hardening
10	Container vulnerabilities	Image scanning, runtime security
11	Supply chain attacks	SBOM, SCA
12	APTs	SIEM, XDR, threat intelligence
13	Lack of visibility	Centralized logging, SIEM
14	Cryptojacking	Cost and resource anomaly detection
15	AI-related attacks	AI security monitoring and guardrails

Cloud Security Posture Management (CSPM)

CSPM continuously identifies and assesses cloud configuration weaknesses.

For example, CSPM can identify:

- Publicly exposed storage.
- Unencrypted databases.
- Unused administrator accounts.
- Unprotected virtual machines.
- Incorrect security group settings.
- Non-compliant cloud resources.

The system can generate alerts and, where safely authorized, automatically remediate misconfigurations.

Major Public Cloud Cybersecurity Incidents Around the Globe

Public cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) have become critical infrastructure for governments, banks,

universities, healthcare organizations, telecommunications companies, and multinational businesses.

However, public cloud environments have also been involved in major cybersecurity incidents, including data breaches, identity compromises, ransomware, misconfiguration, supply-chain attacks, and large-scale service disruptions.

A key distinction is that a cybersecurity incident in a cloud-hosted customer environment does not necessarily mean the cloud provider's underlying infrastructure was compromised. In many cases, attackers exploit customer-side misconfigurations, stolen credentials, vulnerable applications, or insecure access policies.

Below is a selection of significant incidents from different regions and cloud platforms, including their attack mechanisms, consequences, and security lessons.

1. Major global public cloud cybersecurity incidents

1.1 Capital One cloud data breach – USA (2019)

Cloud platform: Amazon Web Services (AWS)

Major data breach

- Approximately 106 million individuals in the USA and Canada were affected.
- Customer information was accessed without authorization.
- The attacker exploited a server-side request forgery (SSRF) vulnerability associated with a web application firewall configuration.

Attack mechanism:

1. The attacker exploited an application-layer vulnerability.
2. The vulnerability enabled access to AWS instance metadata.
3. Temporary AWS credentials were obtained.
4. Excessive permissions enabled access to cloud storage.
5. Customer information was downloaded.

Information exposed:

- Names and addresses.
- Credit scores.
- Credit card application information.
- Social Security numbers for some affected individuals.

Security lessons:

- Properly configure IAM permissions.
- Protect cloud instance metadata.
- Implement least-privilege access.
- Continuously monitor cloud storage access.

- Conduct application security assessments.

1.2 Microsoft Azure and Microsoft cloud identity compromise – Global (2023–2024)

Cloud platform: Microsoft cloud identity infrastructure and Exchange Online

Identity compromise

- The Storm-0558 campaign affected Microsoft customer email accounts.
- Attackers used forged authentication tokens to access certain Exchange Online accounts.
- The incident affected US government and other organizational email accounts.

Attack mechanism:

- Theft of a Microsoft signing key.
- Forging authentication tokens.
- Abuse of cloud identity authentication.
- Unauthorized access to email data.

Security lessons:

- Protect cryptographic signing keys.
- Strengthen cloud identity architecture.
- Monitor authentication tokens.
- Improve logging and detection.
- Apply strict controls to privileged cloud infrastructure.

1.3 Snowflake customer data breaches – Global (2024)

Cloud platform: Snowflake cloud data platform

Large-scale account compromise

- Multiple customer environments were accessed by attackers.
- Stolen credentials were used to access customer accounts.
- Some compromised accounts did not have MFA enabled.

Affected organizations included:

- Ticketmaster.
- Santander Bank.

Attack mechanism:

1. Attackers obtained previously stolen credentials.

2. They authenticated to customer Snowflake environments.
3. They accessed and extracted stored information.
4. Stolen information was used in extortion activities.

Security lessons:

- Enforce MFA.
- Monitor unusual login activities.
- Rotate exposed credentials.
- Implement network access policies.
- Monitor bulk database extraction.

1.4 Microsoft Midnight Blizzard cloud email compromise – USA (2024)

Cloud platform: Microsoft corporate cloud environment

Nation-state intrusion

- Microsoft disclosed that the Russian state-sponsored actor Midnight Blizzard had accessed certain corporate email accounts.
- The attackers used password-spraying techniques against a legacy non-production test tenant.
- They subsequently exploited an OAuth application to access additional corporate email.

Security lessons:

- Secure legacy and test environments.
- Monitor password spraying.
- Audit OAuth application permissions.
- Enforce MFA and privileged access controls.
- Investigate suspicious cloud identity activity.

1.5 Toyota cloud data exposure – Japan (2023)

Cloud environment: Toyota customer cloud environment

Cloud misconfiguration

- Toyota disclosed that vehicle-related customer information had been publicly accessible for an extended period.
- The incident involved cloud configuration errors.
- The company reported that some customer data could have been accessed by unauthorized parties.

Security lessons:

- Continuously audit cloud configurations.
- Restrict public storage access.
- Monitor changes to access permissions.

- Establish automated cloud security posture management.

1.6 European Commission AWS cloud breach – Europe (2026)

Cloud platform: Amazon Web Services (AWS)

Supply-chain compromise

March 2026

- The European Commission disclosed a compromise affecting its public website hosting platform, europa.eu.
- Attackers obtained an AWS secret through the Trivy software supply-chain compromise.
- Approximately 91.7 GB of compressed data was exfiltrated.
- Personal information and email-related data were among the exposed information.
- Data associated with other EU entities may also have been affected.

Attack mechanism:

1. A compromised version of the Trivy security scanning tool entered the software supply chain.
2. AWS credentials were harvested from the affected environment.
3. Attackers used the credentials to access an AWS account.
4. They performed reconnaissance and created additional access credentials.
5. Data was exfiltrated.

Security lessons:

- Secure software development pipelines.
- Protect secrets and API keys.
- Implement short-lived cloud credentials.
- Monitor AWS API activity.
- Apply software supply-chain security controls.
- Use SBOM and dependency verification.

This is particularly important because it demonstrates how a compromised security tool can become a pathway into a major public cloud environment.

1.7 Retool cloud application compromise – USA (2023)

Cloud environment: Google Cloud-hosted infrastructure

MFA and identity attack

- Attackers compromised employee accounts through sophisticated SMS phishing.
- They bypassed the organization's MFA protection by exploiting weaknesses in the authentication process.
- The attackers gained access to internal systems and certain cloud customer environments.

Security lessons:

- Use phishing-resistant MFA.
- Protect employee identity systems.
- Monitor suspicious authentication changes.
- Restrict access to production systems.
- Apply strong session and device security.

1.8 Football Australia AWS data exposure – Australia (2024)

Cloud platform: AWS

Cloud credential exposure

- A security incident involved exposed AWS credentials.
- The incident raised concerns about access to cloud storage and sensitive organizational information.
- It highlighted weaknesses in access-key management and cloud security monitoring.

Security lessons:

- Never expose AWS access keys in source code.
- Rotate compromised credentials.
- Use IAM roles rather than long-lived access keys.
- Monitor S3 and other cloud resource access.

1.9 Retool and other third-party cloud service compromise – USA (2023)

Another important aspect of cloud security is the compromise of services that have privileged access to customer environments.

Sumo Logic AWS credential incident (2023): Sumo Logic disclosed an incident involving unauthorized access to an AWS account and advised customers to rotate certain credentials and secrets.

Security lessons:

- Secure third-party integrations.
- Regularly rotate access keys.
- Restrict vendor permissions.
- Monitor access to secrets and cloud management interfaces.

1.10 Microsoft Azure service disruption – USA (2024)

Cloud platform: Microsoft Azure

Cloud availability incident

18 July 2024

- An Azure infrastructure incident affected virtual machine access in the Central US region.
- Some enterprise customers could not access their hosted services.
- The disruption lasted approximately half a day.

Security lesson: Cloud security must address availability and resilience in addition to confidentiality and integrity. This was an infrastructure/service disruption, not a confirmed malicious cyberattack.

1.11 CrowdStrike global IT outage – Worldwide (2024)

Technology involved: CrowdStrike Falcon and Microsoft Windows

Global operational disruption

19 July 2024

- A defective CrowdStrike software update caused Windows systems to crash.
- Approximately 8.5 million Windows devices were affected.
- Aviation, banking, healthcare, retail, and other sectors experienced disruptions.

Important distinction: **This was not a cloud data breach or a cyberattack. It was a defective security software update with a worldwide operational impact. It shows the systemic risks of relying on centralized software and cloud-connected security ecosystems.**

2. Other significant cloud-related security incidents

The following cases are also useful for understanding different cloud vulnerabilities.

Incident	Year	Cloud environment	Main issue
Accenture cloud storage exposure	2017	AWS S3	Publicly exposed storage
Uber data breach	2016	AWS-related infrastructure	Exposed credentials and unauthorized access
Code Spaces attack	2014	AWS	Compromised AWS control panel and data deletion
Microsoft Cosmos DB (ChaosDB)	2021	Azure	Vulnerability allowing unauthorized database access
DarkBeam data exposure	2023	Cloud-hosted Elasticsearch	Unsecured database
Toyota customer data exposure	2023	Cloud environment	Misconfiguration
Snowflake customer breaches	2024	Snowflake	Stolen credentials and inadequate MFA
European Commission breach	2026	AWS	Software supply-chain compromise

A few additional explanations are particularly important:

A. Microsoft Azure Cosmos DB – ChaosDB (2021)

A vulnerability in Azure Cosmos DB's Jupyter Notebook feature could have allowed unauthorized access to customer database credentials. Microsoft notified affected customers and remediated the vulnerability.

B. Code Spaces – AWS (2014)

An attacker gained control of the company's AWS management console and deleted resources, causing severe business disruption. Code Spaces ultimately ceased operations.

C. Accenture – AWS S3 exposure (2017)

Security researchers discovered unsecured AWS S3 storage containing sensitive corporate information, including credentials and internal data.

These incidents demonstrate that cloud security failures can result in data theft, infrastructure destruction, and even the collapse of a cloud-dependent business.

3. Comparative analysis of major incidents

The following table compares the most important incidents according to their security implications.

Incident	Year	Cloud provider/platform	Primary attack type
Code Spaces	2014	AWS	Account compromise and destructive attack
Accenture	2017	AWS	Public storage exposure
Capital One	2019	AWS	SSRF and IAM exploitation
ChaosDB	2021	Azure	Cloud database vulnerability
Toyota	2023	Cloud environment	Misconfiguration
Storm-0558	2023	Microsoft cloud	Authentication token forgery
Retool	2023	Google Cloud-related services	Spear phishing and MFA compromise
Midnight Blizzard	2024	Microsoft cloud	Password spraying and OAuth abuse
Snowflake	2024	Snowflake	Stolen credentials
CrowdStrike outage	2024	Cloud-connected security ecosystem	Defective software update
European Commission	2026	AWS	Software supply-chain compromise

4. Classification of incidents by cybersecurity vulnerability

An important observation is that different cloud incidents exploit different security weaknesses.

Five particularly important lessons

1. **Identity compromise:** Stolen credentials, weak MFA, and compromised authentication tokens are major attack pathways.
2. **Cloud misconfiguration:** Public storage and incorrectly configured access permissions can expose enormous volumes of data.
3. **Supply-chain vulnerabilities:** A compromised third-party development or security tool can provide access to cloud infrastructure.
4. **Insufficient monitoring:** Organizations must monitor cloud API calls, administrator activities, unusual data transfers, and access-key creation.
5. **Cloud dependency and resilience:** Even when no attacker is involved, a faulty update or infrastructure failure can disrupt critical services worldwide.